

5.1.3 Summation Notation

Consider again the example in which $A_k = 2^k$ represents the number of ancestors a person has in the k th generation back. What is the total number of ancestors for the past six generations? The answer is

$$A_1 + A_2 + A_3 + A_4 + A_5 + A_6 = 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 = 126$$

It is convenient to use a shorthand notation to write such sums. In 1772 the French mathematician Joseph Louis Lagrange introduced the capital greek letter sigma, $\sum$, to denote the word sum (*or summation*), and defined the summation notation as follows:

Definition

If m and n are integers and $m \leq n$, the symbol $\sum_{k=m}^n a_k$ read the **summation from k equals m to n of a-sub-k**, is the sum of all the terms $a_m, a_{m+1}, a_{m+2}, \dots, a_n$. We say that $a_m + a_{m+1} + a_{m+2} + \dots + a_n$ is the **expanded form** of the sum, and we write

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + a_{m+2} + \dots + a_n$$

We call k the **index** of the summation, m the **lower limit** of the summation, and the n the **upper limit** of the summation.